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Chapter 3

Material Planning



Material Requirements Planning Overview

Material requirements planning is a process comprising three functions:

- Run MRP.
- Review MRP.
- Confirm suggested orders.

Run MRP

There are three types of MRP run:

MRP run types	What they do
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Regenerative MRP Run	Examines all current supplies and demands against items designated as MRP items.
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The processing time for a full Material Requirements Planning (MRP) run is such that the run may have to take place over a weekend.

It is suggested that you carry out a regenerative MRP run once a month.

Net Change MRP Run	Considers only those materials that have had any changes to their demand or supply since the last MRP run. This type of run is shorter than a full MRP run and therefore allows you to run MRP more frequently.
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Performing a net change MRP run frequently (e.g. on a daily basis) enables you to take action on demand and supply changes as soon as they happen in order to revise the plan for the materials that are affected.

It is suggested that you carry out a net change MRP run weekly.

Selective MRP Run	Allows you to run MRP for selected items.
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Each type of run generates suggestions for actions which you should carry out in order to satisfy the plan.



Warning: If you carry out a number of net change MRP runs in succession, they will become slower than a regenerative MRP run eventually.

Review MRP

Within Review MRP, you can examine the results from the MRP run in terms of whether the correct orders are in place to meet the plan. You can then action optionally any suggestions which are related to adding, changing, or cancelling orders.

Confirm Suggested Orders

Confirm suggested orders either individually via the review processes or 'en masse' via a standalone routine.

Maintain Model Stockrooms



1/P3M

Use this option to create and subsequently maintain MRP planning models.

You can use the same model for MRP as for MPS. Alternatively, you can have a separate MRP model with different stockrooms and planning periods.

In both cases, having run the option in MPS, you must run the same option in MRP.

This option is the same as a Style Production Scheduling option. For more information, refer to the Maintain Model Stockrooms option in Style Production Scheduling.

Maintain Reporting Profile

**2/P3M**

This option is the same as a Style Production Scheduling option. For more information, refer to the Maintain Reporting Profile option in Production Scheduling.



Note: Having run this option for MPS, you must also run the option again under MRP.

Net Change Reason Codes



3/P3M



Select Net Change Reason Codes on the menu to display the Net Change Reason Codes Maintenance screen.

Use this option to activate or deactivate triggers upon events for a net change MRP run. You do this by maintaining reason codes which control the triggers for certain events.

A net change MRP run is restricted in its update to just certain specified events which have taken place since the last net change MRP run.

Within the manufacturing system, any event that takes place for which you have selected a reason code creates a trigger.

For example, if you select the reason code for Production Order Triggers and subsequently create an unplanned production order, then Style activates a trigger for the next net change MRP run.

When you request a Net Change MRP run, you can view the number of triggers set against each of your selected Reason codes.

Initially, you should run this option to create these reason codes. On subsequent runs, to stop certain transactions from writing triggers, you can run this option again to deactivate the appropriate reason codes.



Warning: Prior to running a net change MRP run for the first time, you must run this option in order to create the reason codes that allow a net change MRP run to take place. The first time you run this program, you must press F8=Update to create the reason codes for the company you are processing. Even if you do not make any changes, ie leave all the codes active, you must still press F8=Update. Until you do so, the net change MRP run ignores all the triggers.



Note: The events listed below activate triggers for MRP controlled items only.

Sales Order Triggers

Because MRP is restricted to planning of materials only, a trigger can be generated only in exceptional cases, ie for sales orders where the ordered item is also an MRP component. Under such circumstances, any of the following events generate a trigger:

- Creation of the sales order.
- Change of date in the sales order.
- Change in quantity in the sales order.

- Additional order lines.
- Deleted order lines.

Unplanned Inventory Transactions Triggers

Sundry issues and sundry receipts force a change in the demand for materials and can therefore generate triggers.

Purchase Order Triggers

If the item ordered is an MRP item, then ad hoc updates fall outside planned changes and affect the expected supply of materials. In this case, any of the following events can generate a trigger.

- Creation of an unplanned purchase order.
- Change of date in an existing purchase order.
- Change in quantity in an existing purchase order.

Production Order Triggers

Ad hoc updates fall outside planned changes and affect the planned demand of the materials attached. Therefore, if an order is for an MRP item, then any of the following events can generate a trigger for the ordered style. If an order is for an MPS item, then any of the following events can generate a trigger for the MRP components only.

- Creation of an unplanned production order.
- Change of date in an existing production order.
- Change in quantity in an existing production order.

Completion of a production order can also generate a trigger for any MRP materials on the order that have an issue quantity less than the required quantity (ie not all the material has been consumed).

Unplanned Issues And Receipts Triggers

You may wish to capture any materials issues that are above the planned quantity, to help you identify and plan for any additional quantity of material required.

You may also wish to capture materials issues that are below the planned quantity, to help you issue the surplus material against another order.

You can capture the following unplanned issues and receipts changes:

- Issue materials. An issue of more materials than required.
- Production booking. Overriding the backflush quantities.
- Production booking. Completing a production order short, ie where the quantity produced is less than the quantity ordered.
- Order completion. Completing or re-opening a production order when the quantity is over the planned quantity.

MR110	ST - Style Test & QA Company	USER QASTYLE	24/06/97
			10:13:53
Net Change Reason Codes Maintenance.			
Code	Description	Active (1=Yes, 0=No)	
01		0	
02		0	
03		0	
04		0	
05	Unplanned Inventory Trans.	1	
06		0	
07	Unplanned Issue or Receipt	1	
08	Purchase Order Maintenance	1	
09	Order Maintenance	1	
10	Sales Order Maintenance	1	+
F3=Exit F8=Update F12=Previous			
08-15	SE	MW	KS IM II KENB20A PET165S2



Note: The reason codes shown on the screen relate to those functions for which you can set triggers at present. Please note that the spare lines are intended for new reason codes that may be included in a future development.

Fields:

Code

This field is for your information only and cannot be amended.

Description

This field describes the group of events triggered by the reason code. You can modify the descriptions (up to 36 characters) for the five reason codes that exist at present.



Note: You can edit the descriptions of the existing reason codes, but you cannot add a new reason code.

Active

This field determines whether or not changes to supply and demand create triggers for the events identified by the reason code. You can enter the following options:

- 0 - No. Do not create triggers for supply and demand changes for this group of events.
- 1 - Yes. Create triggers for supply and demand changes for this group of events.



Note: If you set a reason code here to 1, then you can change this setting to 0 for an individual net change MRP run. This action causes that particular net change MRP run to ignore any triggers set up for the reason code. For more information, see the Net Change Trigger

Selection window in the Run MRP option.

*However, if you set a reason code here to 0,
then no triggers are set up for that reason code.
Consequently, net change MRP runs cannot take
into account any changes for that reason code.*

Functions:

F8

Update. Save the changes you have made to net change reason codes.

Run Material Planning



11/P3M

MRP is a process whose main functions are common to MPS. In summary, MRP:

- Takes all demand on top-level MRP items.
- Compares the demand against the available supply and generates additional suggested demand where appropriate.
- Explodes the supplies into lower level demands.
- Compares the demands against supplies for lower level items.
- And continues down the structures in a similar manner.

The system generates an MRP Reporting Profile report, detailing the run dates and the reporting periods associated with the planning model.

At the end of the run, you can review and action MRP's suggestions.

Prerequisites

For each style that MRP is to consider, you must have defined:

- A planning route for the style under Production Details in the Style/Material Details option.
- A list of materials on the planning route for the style, as defined under the Style Route/BOM option.

In addition, Ensure that you have defined your calendars to cover the duration of the review.



Warning: MPS and MRP are widely-used manufacturing control terms which are used in many different environments.

These terms in Style Production differ from many other uses of the terms in the following way. In Style Production, all manufactured items should be designated MPS items; and all purchased items should be designated MRP items.

The system does not prevent you from contravening this rule. If you do so, the effect will be that the production load imposed by wrongly coded items will not be reflected in the Capacity Review and the recommendations for actions on wrongly coded purchased items will be generated during the MPS run rather than the MRP run.

MRP Selection Screen 1



To display this screen, select Run Material Planning from the menu.

Use this screen to enter a planning model.



Tip: Ensure that you have defined your calendars to cover the duration of the review.



Note: As part of standard processing, the system generates an MRP Reporting Profile report, detailing the run dates and the reporting periods associated with the planning model.



Tip: If the planning route defined for a style has either not been created or been created but has no materials defined, then any supply suggested or existing production orders for such styles are not included within the MRP run. To define the planning route for a style, refer to the Production Details screen under the Style/Material Details option.

Fields:

Planning Model

Enter the planning model to run. To confirm any planning suggestions, this must be the planning model defined on the Company Profile, Planning Options screen.

The stockrooms defined on the selected model determine the scope of the material plan produced. The associated reporting profile defines how information will be presented (that is, bucketed) on reports and enquiries. Only items with defined item/stockroom relationships within the selected model are extracted for planning.



To display MRP Selection screen 2, press **Enter**.

MRP Selection Screen 2



To display this screen, press **Enter** on MRP Selection screen 1.

Use this screen to select the type of MRP run you require and to enter a start date and end date.

Initially, this screen displays the Current Date and End Date from the last MRP run.

MR002	ST - Style Test & QA Company	USER QASTYLE	24/06/97 14:23:27
MRP Selection			
Planning Model		: EF EF model	
Production Calendar		: 01	
MRP Run Type.....		1 = Regenerative 2 = Net Change 3 = Selective	
Include suggested MPS		1 (1=Yes)	
Enter	Current date	030397	
	Overdue days	_____	
	Time Fence days	_____ Use Item Time fence @ (1=Yes)	
	Safety Horizon days	_____	
	End date	300597 Use Item End date @ (1=Yes)	
Suppress low level code generation..		1 (1=Yes)	
F3=Exit F12=Previous F16=Review Selection F8=Start Run F22=Print details			
07-33 S4 MW KS IM II KENB20A PET16SS2			

Fields:

Production Calendar

This field displays, for your information only, the calendar associated with the reporting profile for the selected planning model. This calendar is set up on the Material Requirements Maintain MRP Reporting Profile screen under the Maintain Reporting Profile option within Style Material Requirements Planning.

MRP Run Type

This field indicates the type of MRP run you require. Enter one of the following:

- 1 - Regenerative. Run MRP for all items within the planning model.
- 2 - Net Change. Run MRP only for those items that have had changes to their demand or supply since the last MRP run. Press **F16=Review Selection** to display the Net Change Trigger Selection window, from which you can de-select trigger reason codes.
- 3 - Selective. Run MRP only for those items in the planning model that are within the selection parameters defined. Press **F16=Review Selection** to display the Enter Selection Criteria window, from which you can define your selection.

Include Suggested MPS

This option is only available for a regenerative run. Enter one of the following:

0 - Exclude MPS suggested orders.

1 - Include MPS suggested orders.

Current Date

This field indicates the live date for the plan. The start date and frozen period (Time Fence Days) are calculated from this date. The start date is the first date which is considered for the run. Orders prior to this date are ignored.

The start date is calculated as:

- Current date less overdue days.

Overdue Days

Indicates the number of days of overdue supply and demand you wish to consider.

Accumulated supply and demand in this period is displayed in the OVERDUE column on the MRP Summary Review screen.

Time Fence Days

The period between the current date and the time fence date is regarded as fixed; that is, a frozen period (or schedule) in which no recommendations can be made.

If you wish to incorporate a time fence, you can use this field to enter directly the number of days. Alternatively, you can enter 1 in the Use Item Time Fence field, indicating that you wish to incorporate the production lead time for each style.

Use Item Time Fence

Enter one of the following:

0 - No. Use the global, default Time Fence Days setting.

1 - Yes. Use the production lead time for each style set up under Production Details on the Style/Material Details option.. The production lead time is the 'this level' lead time in days required to produce an item from its immediate components. If the production lead time is not set up for a style, then the MRP run uses the global, default Time Fence Days setting.

Safety Horizon Days

This ensures that the end date is extended so that all generated demand is included in the lower level analysis.

End Date

This field indicates the last date to be considered in the run. To include a safety lead time, enter a value in the Safety Horizon Days field.

Alternatively, use a cumulative lead time for each style plus a safety lead time by entering 1 in the Use Item End Date field on this screen.

Use Item End Date

Enter one of the following:

0 - No. Use the global, default end date.

1 - Yes. Use the cumulative lead time set up under Production Details in the Style Material/Details option. The cumulative lead time is the total lead time in days required to produce an item based on a full explosion of its planning route, including any low level production items included on the route, together with the purchasing lead time of materials and bought out components. If the cumulative lead time is not set up for a style, then the MRP run uses the global, default end date.



Note: For non-production items, the purchasing lead time is used for production and cumulative lead times.

Suppress Low Level Code Generation

This field has a default setting dependent on whether or not the system has detected any changes to planning routes.

If it has, the field defaults to 0 and displays the message 'changes made in structure currently exist'.

If it has not, the field displays 1.

The routine may also be suppressed if the independent Generate Low Level Codes routine is destined to be run prior to running MRP.



Warning: The decision to suppress must only be taken if you are sure that the planning process will not be compromised by the lack of up-to-date data regarding planning routes.

Functions:

F8

Start Run. Submit a batch job to start the MRP run. The system automatically produces a report detailing the parameters set for the run.

F16

Review Selection. If you entered 2 in the MRP Run Type field, then the Net Change Trigger Selection window opens. If you entered 3 in the MRP Run Type field, then the Enter Selection Criteria window opens.

F22

Print Details. Print the run parameters without initiating a run.

Net Change Trigger Selection Window (F16)



To display this window, enter 2 in the MRP Run Type field and select **F16=Review Selection** from MRP Selection screen 2.

Use this screen, for a net change MRP run only, to:

- View the number of unprocessed triggers set by each net change reason code.
- View the total number of styles triggered.
- Select the trigger types that you wish the net change MRP run to process. For example, you can ask the net change run to ignore any triggers set for sales order changes.

The triggers are grouped according to trigger reason codes. You can activate or deactivate triggers for the following reason codes:

- Sundry issues and sundry receipts for MRP items.
- Unplanned issues and receipts.
- Unplanned purchase order updates for MRP items.
- Unplanned production order updates for MRP items and MPS items with MRP components.

Unexpected sales order updates for MRP items.



Warning: Prior to running a net change MRP run for the first time, you must create the reason codes that allow this type of MRP run to take place. To do this, run the Net Change Reason Codes option. Initially, all the codes are marked as active. Please note that even if you do not wish to change the Active status of any of the codes, you must still select F8=Update. If you do not, then the net change MRP run ignores all the triggers.

MR002	ST - Style Test & QA Company	USER QASTYLE	21/07/97 16:29:51
MRP Selection			
Planning Model : ST ST		Production Calendar : 01	

Net Change Trigger Selection		
Sel	Trigger Description	Unprocessed Triggers
1	05 Unplanned Inventory Trans.	0
1	07 Unplanned Issue or Receipt	0
1	08 Purchase Order Maintenance	1
1	09 Order Maintenance	4
1	10 Sales Order Maintenance	0

1=Select	Unprocessed Styles	5
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F3=Exit F12=Previous F18=Select All
F22=Print details

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Fields:

Sel

Select. Select a reason code to make the net change MRP run take into account all triggers covered by the reason code. For each line, you can enter the following option:

1 - Select. Reason code.

To deselect a reason code, press the right **Ctrl** button.



Note: The triggers for one or more reason codes may be inactive and cannot therefore be processed. In this case, the reason code does not appear on this screen. For more information, see the Net Change Reason Codes Maintenance screen.

Trigger Reason

This field is for your information only and cannot be amended.

Description

This field is for your information only and cannot be amended from this window. For more information, see the Description field on the Net Change Reason Codes Maintenance screen.

Unprocessed Triggers

Displays a count of the number of triggers set for each reason code. This field is for your information only and cannot be amended.

Unprocessed Styles

Displays a count of the number of styles that the Net Change run will process for all active reason codes. This field is for your information only and cannot be amended.

Functions:**F18**

Select All. Select all the available reason codes and therefore all the unprocessed triggers. Returns you to MRP Selection screen 2.



To validate your changes and display MRP Selection screen 2, press **Enter** from this window.

Enter Selection Criteria Window (F16)



To display this window, enter 3 in the MRP Run Type field and select **F16=Review Selection** from MRP Selection screen 2.

Use this screen, for a selective MRP run only, to define the scope of the MRP run in terms of the items to be considered.

All the selection parameters on this screen, with the exception of the Value/Usage Class field, are defined for each style via the Style/Material Details option in Production Definition Management. Value/usage classes relate to specific stockrooms and are defined within Style Inventory Management.



Note: If you intend to process all MRP items, then carry out a regenerative MRP run instead by entering 1 in the MRP Run Type field on MRP Selection screen 2.

MR002 ZS - Style Demonstration Co. USER GBJBAMS 15/04/96
15:25:48

MRP Selection

Planning Model : ZS
Production Calendar : 01

MRP Run Type..... 2 1 = Regenerative

Enter Selection Criteria

Item.....?.. DAPSINGLE to FB870

Planner.....?.. to

Item Group.....?.. to

Value/Usage Class.... to

Preferred Supplier.... to

GT Family..... to

F4=Prompt F12=Previous

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Fields:

Style

Enter a range of styles. To include all styles, leave this field blank.

Planner

Enter a range of planners. To include all planners, leave this field blank.

Style Group

Enter a range of Product Group Minor Codes. To include all Product Group Minor Codes, leave this field blank.

Product Group Minor Codes are used to group styles into product families for sales and stock forecasting, but can be used for other purposes if product families are not required.

Value/Usage Class

Enter a range of Value/Usage Classes. To include all Value/Usage Classes, leave this field blank..

Preferred Supplier Range

Enter a range of preferred suppliers. To include all preferred suppliers, leave this field blank.

GT Family

Enter a range of Fabric Types. To include all Fabric Types, leave this field blank.

Confirm Suggested Orders



12/P3M

Use this option to confirm selected or all suggested orders. Once confirmed, the orders are taken into account by any future MRP runs.



Note: If you have set the Allocation Of Materials At field to 1 (ie allocate at order confirmation) on Maintain Company Profile screen 4, then material for the production orders is allocated automatically upon confirmation.



Warning: If you have coded your items as recommended, that is, manufactured items as MPS and purchased items as MRP, you should not need to use this screen from within MRP.

MRP Confirmation Of Suggested Orders Screen



To display this screen, select **Confirm Suggested Orders** from the menu.

Use this screen to select, optionally, a range of dates and planners.

```
MR052                ZS - Style Demonstration Co.                USER GBJBAMS        15/04/96
                                                                15:39:16
MRP - Confirmation of Suggested Orders

                        Model... ZS

                        From Start Date.. █_____ To Start Date.. _____
                        From Planner...?.. ____ To Planner...?.. ____

F3=Exit  F4=Prompt

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```

Fields:

Model

This field displays, for your information only, the 'live' planning model and description, defined on the Company Profile, Planning Options screen.

From/To Start Date

Enter a range of dates to include only production orders with a start date within the selected range. To include all orders, leave this field blank.

Any dates entered must be within the date range of the last MRP run.

Planner Range

Enter a range of planners to select only orders associated with items defined with the entered planner codes, or leave blank to include all planners.



To display the Confirm Orders screen, press **Enter**.

Confirm Orders Screen



To display this screen, press **Enter** from the MRP Confirmation Of Suggested Orders screen.

Use this screen to select which suggested orders to confirm and to submit a batch job to mark all selected orders as confirmed orders..

This screen displays all suggested production orders that meet the selection parameters entered on the MRP Confirmation Of Suggested Orders screen. Initially, each order is set for selection. You can reverse this setting for individual suggested orders.

Fields:

O

Option. Enter one of the following.

0 (Or delete entry) - Prevent the order from being processed.

1 - Select an order for processing; this is the default setting displayed against each selected order.

Functions:

F8

Confirm all selected production orders.

Orders with a flag setting of 1(create order) are processed. Confirmed orders for the suggested due dates and quantities are generated. The function uses the designated planning route entered on the item definition, Production Details screen to generate operation and material requirement details.

The status of the order is changed to 'Confirmed'.

Review



13/P3M

Use this option to review the results of an MRP run. For a live planning model, you can create and maintain actual supply.

Material Requirements Planning Enquiry Screen



To display this screen, select Review from the menu.

Use this screen to enter a planning model.

Fields:

Planning Model

Enter the planning model you wish to review.



To display the Enquiry screen, press **Enter**.

Enquiry Screen



To display this screen, press **Enter** from the Material Requirements Planning Enquiry screen.

Use this screen to enter, optionally, a material, a planner and a date range.

Fields:

Select Item

Enter a material (optional).

Planner

Enter a Planner, as set up under Production Details within the Style/Material Details option (optional).

Date Last Planned From / To

To display materials that were last planned within a specific date range, enter a range of dates (optional).

The default is the run date of the last MRP process.

To select all materials for display, leave both this field and the Select Item field blank.

If you enter a style or material in the Select Item field, leave this field blank.



To display Enquiry Detail Screen 1, press **Enter**.
If the selected material has no detail available,
then the Enquiry Summary screen displays.

Enquiry Detail Screen 1



To display this screen, press **Enter** from the Enquiry screen.

This screen displays a summary of all supply and demand details associated with a material, including recommendations and time-projected inventory balances.

From this screen, you can display other screens which give you more detail and from which you can confirm suggested actions. For a more detailed breakdown down to variant level, select option 7 against a supply line.

Using **F21=Change Seq**, you can display data in either of two sequences:

- By MRP recommended due date. These are dates incorporating any suggested changes made by MRP.
- By current due date. These are the dates which currently exist and which the MRP run may have recommended for change.

If a production order suggestion exceeds the maximum order quantity defined for a material, then this screen highlights the production order in reverse video.

Stock Forecasts

Stock forecasts are not subtracted as demand but there may be a corresponding supply order which is scheduled to provision the stock level defined in the forecast. This stock is additional to safety stock, so the quantity available will reflect stock forecast demand.

Stock forecasts should be viewed as an additional hedging technique, with the flexibility to amend the desired stock level by week. Additional stock forecasts will cause an additional level of stock to be planned.

MR350	ZS - Style Demonstration Co.				USER GBJBAMS	15/04/96	
						15:30:05	
Enquiry - Detail		Planning Model		ZS			
		Start 11/08/95		Time Fence 22/05/95		End 28/07/95	
Item.... FAB1		Fabric				MPS Item 0	
MRP <---- Demand -----> <----- Supply ----->							
0	Date	Order	Quantity	St	Order	Quantity	St Action Available
■	9/05		45	SW			78
-	23/05	W000417	35	RW			43
-	24/05	W000437	35	RW	32	SW	40
-	24/05	W000440	35	RW			5
-	25/05	W000452	35	RW			30-
-	26/05	W000463	35	RW	15	SW	50-
-	30/05	W000503	35	RW	35	SW	50-
-	1/06	W000578	18	RW	18	SW	50-
-	2/06	W000409	325	RW	1602	SW	1226
-	2/06	W000412	35	RW			1191
-	2/06	W000413	419	RW			772
-	2/06	W000415	35	RW			73 +
5=Maintain Supply 6=Peg Demand 7=Supply Breakdown							
F3=Exit F5=Refresh F12=Previous F17=Summary F18=Header Details							
F19/F20=Next Item/Planner F16=Prev.Item F21=Change Sequence F22=Detail Line							
09-02		SP	MW	KS	IM	II	KENB20A PET166S2

Fields:

MPS Item

This field displays, for your information only, the value set up under Planning Type under Production Details within the Style/Material Details option. 0 indicates MRP. 1 indicates MPS.

O

Option. Enter one of the following:

5 - Maintain Supply. This option is only available for the live planning model. Enter the option as follows:

- Against an existing production order to amend details.
- Against a suggested order to create a Planned or Confirmed status order. Alternatively, to confirm multiple orders rather than selecting individually via this review screen, use the Confirm Suggested Orders option.
- Against a suggested purchase order to modify the order. Opens the Purchase Order Maintenance window.

6 - Peg Demand. Display the parent that generated a particular demand, using Enquiry Detail Screen 2. The demand item must be a material requirement on a production style and have a low level code of 1 or greater. After viewing any number of levels of parent details, **F12=Previous** displays the originating item in a chain of pegging requests.

7 - Supply Breakdown. Open the Suggested W/O window, displaying the variant breakdown of the material under review. No changes can be made to the quantities displayed. If a material has no variants, then the screen does not display this option.



Note: Lines without the option to amend indicate an accumulated demand/supply on the first item.

Date

This field displays, for your information only, one of the following:

- For dependent demand, this is the required date of the item, and reflects a higher level MPS demand.
- For independent demand, this is the forecast due date, or sales order shipment date.
- For a supply, it is one of the following, depending on the presentation sequence selected (see **F21=Change Seq**). For supplies with no recommended date change the sequence remains the same.
 - MRP recommended due date.
 - Current date of the order.
- 99/99 indicates a zero demand with an existing supply.

Demand

St

Demand Status. The following status codes can be displayed:

Demand Status Codes

CA	Cancelled
CD	Cumulative demand, if cumulative consumption of forecast
CW	Confirmed production order
FC	Sales forecast
FS	Stock forecast
FX	Forecast (excluded)
IW	Active production order
MA	[MPS] Manual adjustment
MX	[MPS] Manual adjustment (excluded)
PW	Planned production order
PX	Suggested purchase order (excluded)
RW	Released production order
SO	Sales order
SP	Suggested purchase order
SW	Suggested production order
SX	Sales order (excluded)
WX	Production order (excluded)
YF	STYLE sales forecast
YP	STYLE suggested purchase order
YS	STYLE stock forecast
YW	STYLE suggested production order

Supply

St

Supply Status. The following status codes can be displayed:

Supply Status Codes	
CN	Cancelled
CO	Complete production order
CP	Confirmed purchase
CW	Confirmed production order
GI	Goods inwards
IS	In inspection
IW	Active production order
PW	Planned production order
RW	Released production order
SP	Suggested purchase order
SW	Suggested production order
YW	STYLE suggested production order
YP	STYLE suggested purchase order

Action

Action Code. The recommended action for the supply order.

The advice is based on the latest status of material requirements and the current inventory order status. The following recommendations can be made:

Action Codes	
CA	Cancel order
CQ	Change quantity (mix)
EF	Check effectivity
EI	Reschedule 'in' and check effectivity
EICQ	Reschedule 'in' and change quantity and check effectivity
EO	Reschedule 'out' and check effectivity
EOCQ	Reschedule 'out' and change quantity and check effectivity
EQCQ	Change quantity and check effectivity
RI	Reschedule 'in'
RICQ	Reschedule 'in' and change quantity
RO	Reschedule 'out'
ROCQ	Reschedule 'out' and change quantity

Available

This field displays, for your information only, one of the following according to the sequence selected (see **F21=Change Seq**):

- **MRP Recommended Due Date.** This field indicates the planned available stock and reflects MRP supply recommendations to meet demand.
 - **Planned Available Stock.** Physical stock (first detail line only) or quantity available on previous date less demand for that date less safety stock plus supplies recommended on that date.
 - **MRP Supply Recommendations.** Since MRP aims to protect the defined safety stock level of the item, the balance normally shows a quantity equivalent to safety stock.
- **Current Due Date.** This field indicates the current supply schedule in terms of quantities and due dates and the resultant availability against demand. It is calculated for each day that a demand or supply line is present and represents the style phased projected inventory level.
- **Physical stock (first detail line only)** or quantity available on previous day less demand for date less safety stock plus supplies currently due on date.



Note: Dependent demand for a given date is always generated from higher level recommendations and represents the MRP suggested schedule, and does not change when the supply sequence is changed

Functions:

F5

Refresh. Refresh screen and redisplay data following manual amendments

F16

Prev. Item. Display previous material.

F17

Summary. Display the Enquiry Summary screen.

F18

Header Details. Display the Header Details screen.

F19

Next Item. Display the next material.

F20

Planner. Display the next planner in the sequence.

F21

Change Seq. Toggle between displaying details in MPS recommended due date sequence and current due date sequence.

F22

Detail Line. Shows additional demand information for each supply and demand line, as follows.

- Additional Demand Information. Shows the style that is creating a particular generated demand for a material.
- Additional Supply Information. Shows the current due date of the order and the earliest and latest start date. For purchased orders, the receipt date may be the due date, promised date or expected date, whichever is the greatest.

To toggle between the two sequences, select **F21=Change Seq.**

Production Order Maintenance screen (Option 5)



To display this screen, enter 5 in the Option field against a suggested production order on Enquiry Detail screen 1.

Use this screen to create or maintain a production order. In particular, you can convert a suggested order into a planned order.

From this screen, you can maintain production orders at suggested, planned, confirmed or released status. However, within the context of planning, you are more likely to change the status of the order either from suggested to planned, or from planned to confirmed.

For more information on maintaining confirmed orders, refer to Style Production Control.

This screen is described fully under the Maintain Orders option in Style Production Control. For more details, refer to Production Order Maintenance screen 2.



Tip: If you have coded your manufactured items as MPS and purchased items as MRP, as recommended, then you should not need to use this screen from within the Review option.



Note: This option is only available for the 'live' planning model.



Note: If the Order Maintenance reason code is active, then the following events create a trigger for the next MRP run:

- Creation of a production order.
- Change of date in an existing production order.
- Change in quantity in an existing production order.

For more information about reason codes, see the Net Change Reason Codes Maintenance screen.

PC151	ZS - Style Demonstration Co.	USER GBJBAMS	15/04/96
Production Order Maintenance			15:31:06
			Week: 16 96
Production Order : _____		* Enter new order details.	
Order Type.....: 1			
Style: FAB1			
Route Code....?..: _			
Quantity.....: 32		UOM.....:	
Due Date.....: 240595		Stockroom:	
Start Date.....: _____		Due Week.....:	
		Start Week.....:	
Calendar Code...?: _		Production Stage.: _	
Planning Filter.?: _			
Print Order.....: 0 (1=Yes)		Reference 1 _____	
		Reference 2 _____	
F3=Exit F4=Prompt F8=Add new order F12=Previous F16=Item search F19=Confirm			
08-22	Sf	MW	KS
		IM	II
		KENB20A	PET166S2

Purchase Order Maintenance window (Option 5)



To display this window, enter 5 in the Option field against a suggested, planned or confirmed purchase order on Enquiry Detail screen 1.

Use this screen to maintain the due date and/or quantity against a suggested purchase order.



Note: This option is only available for the live planning model.

MR350	RK - Style Production Company	USER GBJBARK	15/10/96 14:00:42
Enquiry - Detail		Planning Model CM Central Model	
Item.... DENIM		Start 14/10/96 Time Fence 15/10/96 End 12/12/96	MPS Item 0
Purchase Order Maintenance Item: DENIM Denim Due Date: 15/10/96 Quantity: 3000.000 F5=Refresh F12=Previous			
MRP <--- Demand ---> <-----> 0 Date Order Quantity St Order 15/10 <u>5</u> 15/10			
S=Maintain Supply 6=Peg Demand 7=Supply Breakdown F3=Exit F5=Refresh F12=Previous F17=Summary F18=Header Details F19/F20=Next Item/Planner F16=Prev.Item F21=Change Sequence F22=Detail Line 11-51 SE MK KS IM II KENB20A PET166S3			

Fields:

Due Date

Maintain due date.

Quantity

Maintain quantity.

Functions:

F5

Refresh. Restore the date and quantity to the values displayed when the order was selected.

Enquiry Detail Screen 2 (Option 6)



To display this screen, enter 6 against a demand item in the Option field on Enquiry Detail screen 1.

Use this screen to view the detailed production schedule for the parent that is causing a generated demand on a material.

The demand is shown together with its own MPS supply/demand details.

If the material is used for more than one style, you can page through each style using **F19=Next Item**.



Note: This option only applies to materials.

MR350	ZS - Style Demonstration Co.	USER GBJBAMS	15/04/96
			15:34:35
Enquiry - Detail	Planning Model ZS		
Item.... FB070	Start 11/08/95	Time Fence 22/05/95	End 28/07/95
	White Blouse		MPS Item 1
MRP <---- Demand -----> <----- Supply ----->			
0	Date Order	Quantity St Order	Quantity St Action Available
22/05			18 SW 162
22/05		W000409	120 RW RI 22/05/95 202
22/05		W000412	12 RW RI 22/05/95 294
22/05		W000413	156 RW RI 22/05/95 450
22/05		W000415	12 RW RI 22/05/95 462
22/05		W000416	18 RW RI 22/05/95 480
22/05		W000417	12 RW RI 22/05/95 492
22/05		W000418	12 RW RI 22/05/95 504
22/05		W000419	12 RW RI 22/05/95 516
22/05		W000420	12 RW RI 22/05/95 528
22/05		W000422	10 RW RI 22/05/95 538
22/05		W000423	10 RW RI 22/05/95 54 +
S=Maintain Supply 6=Peg Demand 7=Supply Breakdown			
F3=Exit F5=Refresh F12=Previous F17=Summary F18=Header Details			
F19/F20=Next Item/Planner F16=Prev.Item F21=Change Sequence F22=Detail Line			
09-02	SP	MW	KS IM II KENB20A PET166S2

Suggested W/O Window (Option 7)

- ➡ To display this window, enter 7 against a supply line in the Option field from Enquiry Detail Screen 1.

This window shows the variant style/colour or style/colour/size distribution of the order. The review is for display only.

MR350

ZS - Style Demonstration Co.

USER GBJBAMS

15/04/96
15:35:13

Enquiry - Detail

Planning Model ZS

Start 11/08/95 Time Fence 22/05/95 End 28/07/95

Item.... FAB1 Fabric MPS Item 0

MRP <---- Demand -----> <----- Supply ----->

0 Date Order Quantity St Order Quantity St Action Available

FAB1

Colour

White

Fabric

█ 18

More:

Suggested W/O

18

F12=Previous

F24=More keys

12-19

SA

MW

KS

IM

II

KENB20A

PET166S2

Enquiry Summary Screen (F17)



To display this screen, select **F17=Summary** on Enquiry Detail screen 1.

This screen displays, for your information only, a summary of the item supply and demand schedule, using the reporting periods defined for the planning model via the Maintain Reporting Profile option. The screen also shows inventory and production details defined for the item.

To display next and previous period details respectively, use the PgUp and PgDn keys.

MR350	RK - Style Production Company		USER GBJBARK	15/10/96	14:01:48
Enquiry - Summary		Planning Model	CM Central Model		
		Start 14/10/96	Time Fence 15/10/96	End 12/12/96	
Item.... DENIM		Denim		MPS Item 0	
Planner.		Product Group..	UOM MT		
GT Family.....			Last Planned....	15/10/96	
Order Policy.. H	Opening Stock.	800.000	Fixed Qty...	.000	
Type..... P	Safety Stock.	1000.000	Min.Batch...	1000.000	
Production Seq	Avg.Wkly Usg..	.000	Max.Batch...	.000	
Lead Time.....	Supplier.....		Multiples...	500.000	
	<u>OVERDUE</u>	<u>15/10/96</u>	<u>12/11/96</u>	<u>10/12/96</u>	<u>TOTAL</u>
Actual Dem.					
Forecast...					
Adj.Demand.					
Supply.....		3000			3000
Available..	800	3000	3000	3000	
Net Reqmt..	200	2000-	2000-	2000-	
Plan Recpt.		4000			4000
Plan Avail.	800	4000	4000	4000	
F3=Exit	F12=Previous	F17=Detail	F19=Next Item	F20=Next Planner	
			F16=Previous Item		
10-02	SP	ML	KS	IM	II KENB20A PET166S3

Fields:

Opening Stock

Displays the current inventory stocking level. Calculated as:

- Physical stock less Frozen stock.

Overdue

First column. Displays demand and supply falling in the overdue period defined by the Overdue Days field on the MRP Run Parameters screen. The period is defined as:

- Current date less Overdue days.

Actual Dem.

Actual demand. Calculated as:

- Total sales demand plus Dependent demand due in period.

Where Dependent demand is material requirements for orders offset by the lead time from the parent order due date.

Forecast

Displays the sales forecast in the period.

Adj. Demand

Adjusted demand. Displays the calculated demand due in the period. This quantity is potentially determined by three factors:

- Demand policy for the item, set on the Production Details screen.
- Forecast consumption method defined on the Company Profile, Planning Options screen.
- Manual adjustments made to demand via the Review Demand option.

Supply

Displays the total of existing supplies in the period.

Available

Displays the available quantity in the period. Calculated as:

- Prior period quantity available plus Supply less Adjusted demand.

In the first period, prior available is equal to Opening Stock.

Net Reqmt

Net requirement. Calculated as:

- Adjusted demand plus Safety stock plus Largest stock forecast in period less Prior period available less Existing supplies.

Plan Recpt

Planned receipt. Displays the total of all MRP recommended supplies (production and purchase orders) due in the period.

Plan Avail.

Planned available. Calculated as:

- Prior period planned available plus Planned receipts less Adjusted demand.

In the first period, previous period planned available is equal to the Opening Stock balance.

Total

The last column shows the sum of all periods for each of the rows except Available, Net Requirement, and Planned Available.

Header Details Window (F18)



To display this window, select **F18=Header Details** from Enquiry Detail screen 1.

This window displays, for information only, inventory and production details for the selected parent style.

MR350	2S - Style Demonstration Co.	USER GBJBAMS	15/04/96
			15:37:50
Enquiry - Detail	Planning Model 2S		
	Start 11/08/95	Time Fence 22/05/95	End 28/07/95
Item.... FAB1	Fabric		MPS Item 0
MRP <---- Demand -----> <----- Supply ----->			
0	Date Order	Quantity St Order	Quantity St Action Available
Header Details			
Planner.....	Item Group.....	Date Last Planned...	2/06/95
UOM..... MT	Phy. Stk..	122,500 Fixed Qty..	.000
Order Policy A	Safety Stk	.000 Min. Qty...	.000
Item Type... M	Lead Time. .00	Max. Qty...	.000
Supplier....	Value/Usage Class.. 0	Multiples..	.000
GT Family... MAIN MATERIAL G			
F12=Previous			
-	2/06 W000413	419 RW	772
-	2/06 W000415	35 RW	73 +
5=Maintain Supply 6=Peg Demand 7=Supply Breakdown			
F3=Exit	F5=Refresh	F12=Previous	F17=Summary
F19/F20=Next Item/Planner	F16=Prev.Item	F21=Change Sequence	F22=Detail Line
16-02	SE	MW	KS
		IM	II
		KENB20A	PET166S2

Enquiry Detail Screen 3 (F22)



To display this screen, select **F22=Detail Line** from Enquiry Detail screen 1.

This screen displays an additional line of information beneath each supply/demand detail.

The Order column displays the order line number of the purchase order. If the purchase order is a blanket order, the order line number is followed by / and the blanket sequence number.

Options and functions are the same as Enquiry Detail screen 1.

MR350	ST - Style Test & QA Company			USER QASTYLE	9/04/98
					9:42:49
Enquiry - Detail		Planning Model	SM Stuart's Test Model		
		Start 30/11/97	Time Fence	2/12/97	End 31/12/01
Item.... SAM/PURP		test			MPS Item 0
MRP <---- Demand -----> <----- Supply ----->					
0	Date Order	Quantity	St Order	Quantity	St Action Available
-	10/08	90	FC P011707	90	CP 0
-	98		004/	Due 10/08/98	
-	7/09	90	FC P011707	90	CP 0
-	98		005/	Due 7/09/98	
-	28/09	90	FC P011725	90	CP RI 28/09/98 0
-	98		001/00001	Due 19/10/98	
-	19/10	90	FC P011725	90	CP RI 19/10/98 0
-	98		001/00002	Due 16/11/98	
-	16/11	90	FC	90	SP 0
-	98			Due 16/11/98 Es 9/11/98 Ls 9/11/98	
-	7/12	90	FC	90	SP 0
-	98			Due 7/12/98 Es 30/11/98 Ls 30/11/	
5=Maintain Supply 6=Peg Demand 7=Supply Breakdown					
F3=Exit F5=Refresh F12=Previous F17=Summary F18=Header Details					
F16=Prev.Item F19=Next Item F20=Planner F21=Change Seq F22=Detail Line					
MA	a				09/002

Fields:

(Item Code & Description)

For dependent demand, the system displays the parent item (code) and description originating the demand.

Due

Current date for supply.

Es

Earliest start date.

Ls

Latest start date.

(Receipt Date)

For purchase orders, the receipt date is the latest of due date, promised date or expected date.

Material Plan Report



21/P3M

This report produces a full listing of the material requirements plan for review and evaluation.

MRP Schedule Report Screen

 To display this screen, select Material Plan Report from the menu.

Use this screen to select a planning model and a planning date range.

MR321P ZS - Style Demonstration Co. USER GBJBAMS 15/04/96
15:46:13
MRP Schedule Report

Model...?.. █

Date Last Planned..... 15/04/96 To 15/04/96

Print Items With No Details.. _ (1=Yes)

F3=Exit F4=Prompt F14=Additional Selection
06-29 **S** MW **K** IM II KENB20A PET166S2

Fields:

Model

Enter the planning model to be reported.

Date Last Planned

Only items which have been planned during the date range entered will be selected for inclusion on the report.

Print Items With No Details

You can decide whether to include items within the planning model for which no supply or demand has been recorded. Enter one of the following:

0 - No.

1 - Yes.

Functions:

F14

Additional Selection. Opens the Enter Selection Criteria window, from which you can select a range of styles on which to report.

 To generate a Material Plan report, press **Enter** from the MRP Schedule Report screen.

Enter Selection Criteria Window (F14)

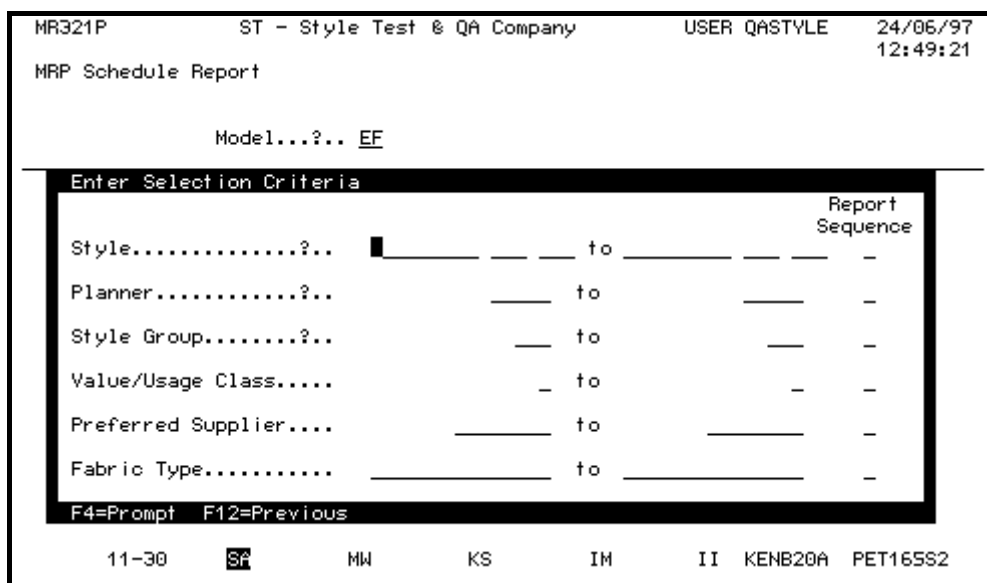
 To display this window, select **F14=Additional Selection** from the MRP Schedule Report screen.

 To display this window, select **F14=Additional Selection** from the MRP Planner Action Report screen.

Use this window to select a range of styles for inclusion in the report. The window offers further selection options based on parameters set on the item's definition, including planner code, preferred supplier etc.

Enter ranges of items to include on the report. The report only prints an item if the item matches all the selection parameters you enter. If you make no selections, then the report lists all items.

All selection parameters excluding value/usage class are defined for an item via the Style/Material Details option (1/P1M) in Production Definition Management. Value/usage classes are defined within the Style Inventory Management application, and relate to specific stockrooms.



The screenshot shows a terminal window titled "MRP Schedule Report" with a header containing "MR321P", "ST - Style Test & QA Company", "USER QASTYLE", and a date/time stamp "24/06/97 12:49:21". Below the header, the text "Model...?.. EF" is displayed. The main area is titled "Enter Selection Criteria" and contains a table with the following fields and headers:

				Report Sequence
Style.....?..		to		-
Planner.....?..		to		-
Style Group.....?..		to		-
Value/Usage Class.....		to		-
Preferred Supplier....		to		-
Fabric Type.....		to		-

At the bottom of the window, there are function key prompts: "F4=Prompt F12=Previous" and a status bar with the text "11-30 SA MW KS IM II KENB20A PET16SS2".

Fields:

Report Sequence

You can sequence your report by entering up to three sequence fields. Enter 1 against the primary report sequence, 2 and 3 against the minor sequences.

For example, you can produce a report showing items in value/usage class sequence within planner responsibility.

Style

Enter a range of item codes or leave blank to include all items.

Planner

Enter a range of planners or leave blank to include all planners.

Style Group

Enter a range of product group minor codes or leave blank to include all product group minor codes.

Product group minor codes are used to group styles into product families for sales and stock forecasting; but could be used for other purposes if product families are not required.

Value/Usage Class

Enter a range of value/usage classes to include or leave blank to include all value/usage classes.

Preferred Supplier

Enter a range of preferred suppliers or leave blank to include all preferred suppliers.

Fabric Type

Enter a range of fabric types or leave blank to include all fabric types.



Press **Enter** on the Enter Selection Criteria window to display the previous screen.

Recommendation Report



22/P3M

This report produces a summarised action list of recommended changes to supply in the planning model. The report can also identify overdue supplies, that is, production orders with a due date earlier than the current date.

MRP Planner Action Report Screen



To display this screen, select Recommendation Report from the menu.

```
MR344                ZS - Style Demonstration Co.                USER GBJBAMS        15/04/96
                                                                15:46:26

MRP Planner Action Report

Model...?... __

Planner Action Review..... 15/04/96 To 15/04/96

Include Overdue Orders..... _ (1=Yes)

Detail Level ..... 1 (1=Style / Material Group)
                    (2=Full Product)

F3=Exit   F4=Prompt   F14=Additional Selection

06-31    S2    MW    KS    IM    II  KENB20A  PET166S2
```

Fields:

Model

Enter the planning model to be reported.

Planner Action Review

This is established by review period. Enter the earliest and latest start dates to consider for MRP supply exceptions.

Include Overdue Orders

You can decide whether to include details of orders with a due date earlier than the current date. Enter one of the following:

0 - No.

1 - Yes.

Detail Level

Enter one of the following:

1 - Style/Material Group. Report at style/material group level.

2 - Full Product. Report at full product detail level 2. A matrix is prompted beneath the order header information to show the breakdown into the variants.

Functions:

F14

Additional Selection. Display the Enter Selection Criteria window, from which you can select a range of styles on which to report. This window is described under the Material Plan Report option.

Valuation Report

1...
2...

23/P3M



To display the MRP Valuation Report screen, select Valuation Report from the menu.

This report assesses the inventory and purchase order commitment of a proposed material requirements plan in standard or current cost terms. The selected plan is translated from a quantitative plan by extending quantities by the unit costs.

```

MR022                ZS - Style Demonstration Co.                USER GBJBAMS                15/04/96
                                                                15:46:37

MRP - Valuation Report

Model?... █          Start      Current      Time Fence      End
                   0/00/00      0/00/00      0/00/00      0/00/00

-----
Basis..... _
      3=One Thousands
      4=Ten Thousands
      5=Hundred Thousands
      6=Millions

1=Summary   2=Detail..... _
1=Std Cost  2=Cost Set..... _
Cost Set..... _____
Select MRP  Item Types..... _
      1=Manufactured
      2=Purchased
      3=Both

-----
F3=Exit    F4=Prompt    F8=Submit job

      06-12    SR        MW        KS        IM        II    KENB20A    PET166S2

```

Fields:

Model

Enter the planning model to be reported.

Basis

This indicates the basis for 'rounding' cost values. Detail calculations will take place prior to rounding on the report.

Summary Or Detail

Enter one of the following:

- 1 - Summary. Print calculated total model information only.
2 - Detail. Include detail item values.

Standard Cost Or Cost Set

Enter one of the following:

- 1 - Std Cost. Carry out costing using standard costs.
- 2 - Cost Set. Carry out costing from the cost set that you specify in the Cost Set field.

Cost Set

If you selected 2 (Cost Set) in the Standard Cost Or Cost Set field, enter the cost set on which calculations are to be based.

Select MRP Item Types

Enter one of the following:

- 1 - Manufactured. Include production items only, item type M (production).
- 2 - Purchased. Include purchased items only, item types P (purchased materials), B (bought out components) and C (consumable tools).
- 3 - Both. Include all items.

Functions:

F8

Submit Job. Submit a batch job to generate the



Select **F8=** on the MRP Valuation Report screen to generate the Valuation report.

Reporting Profile

**24/P3M**

To display the Material Requirements Print MRP Reporting Profile screen, select Reporting Profile from the menu.

This option is the same as a Style Production Scheduling option (34/P2M). For more information, see Reporting Profile Report in Production Scheduling.

MRP Log History



25/P3M



To display the MRP Log History Selection screen, select MRP Log History.

This report provides MRP run information to help you decide whether to carry out a net change MRP run or a full MRP run. Depending upon the number of styles processed, it may be more efficient to perform a full regeneration rather than a net change.

To help you make this decision, this report provides the actual times of MRP runs and their modes.

The report captures the following MRP run information:

- Start date.
- Start time.
- End date.
- End time.
- Run time.
- Run type.
- Styles triggered.
- MRP styles processed.
- MPS styles processed.
- Total styles processed.

MR063	ST - Style Test & QA Company	User QASTYLE	24/06/97
			10:18:47
MRP Log History Selection			
Planning Model.....?.			
From Date..... 0/00/00			
F3=Exit F4=Prompt			
06-30	SP	MW	KS IM II KENB20A PET16SS2

Fields:

Planning Model

Enter the planning model to be reported.

From Date

Enter the date from which to commence the report.



Press **Enter** to validate the data you have entered, generate the MRP Log History report and return to the menu.